ATHUL TITUS

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SUMMARY

Third-year Computer Science (Data Science) student with experience in building web and mobile applications, participating in hackathons and innovation-driven project events, and applying structured approaches to real-world problem solving. Demonstrates the ability to work effectively in teams, communicate ideas clearly, and adapt to new technologies through continuous learning and hands-on implementation. Interested in developing practical, scalable software solutions and strengthening analytical, backend, and AI integration skills.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, HTML/C
Databases: MySQL
Tools: Git, VS Code, Sublime Text
core skills: Problem-solving , Team collaboration , Leadership

PROJECTS

Cymonic — AI-Powered Smart Talent Selection Engine
Web Application — React, FastAPI, NVIDIA NIM, Llama 3

- Engineered a full-stack AI recruitment platform that replaces keyword-matching ATS systems with semantic intent analysis, leveraging Meta's Llama 3 (8B & 70B) models via NVIDIA NIM endpoints.
- Built a multi-format resume ingestion pipeline (PDF, DOCX, Images) using OCR parsing and LLM-based JSON structuring, with real-time SSE progress streaming to a React/Vite frontend.
- Implemented a weighted semantic scoring engine with recruiter-adjustable parameters, an anti-fraud credibility detector for keyword-stuffing detection, and an AI-driven interview question generator — deployed on Vercel and Render.

Microplastic Analyzer
Data Science Project — Python, K-Means, Apriori, Data Mining

- Analyzed microplastics in food products using data mining and machine learning techniques.
- Applied K-means clustering to group samples based on contamination characteristics.
- Used the Apriori algorithm to discover frequent patterns and associations in microplastic presence.
- Interpreted results to identify contamination trends and potential risk patterns.

BB84 QKD Simulator
Mini-Project — TypeScript, Python, Qiskit, React, Flask, Cryptography

- Developed an end-to-end BB84 Quantum Key Distribution simulator with Recursive BB84 protocol, rolling bias, and Cascade error reconciliation.
- Engineered secure P2P messaging using dynamically generated quantum keys and smart abort classification.
- Built scalable web architecture integrating Qiskit (quantum computing), Flask (API backend), and React (frontend).
- Enabled advanced cryptographic functionality for secure communication channels.

EDUCATION

Mar Athanasius College of Engineering, Kothamangalam
Bachelor of Technology in Computer Science and Engineering (Data Science)

VOLUNTEERING EXPERIENCE

Program and Technical Team member – IEEE MACE and google Develepor Group On Campus *Feb 2025- Present*
Organising Team member- AISA MACE (Space Club) *June 2025-Present*
.hack() – Logistics Team, IEEE MACE
Lumora(Designathon) – Logistics and Hospitality Lead , GDG On Campus MACE

CERTIFICATIONS

- NPTEL – Introduction To IOT *Jul – Oct 2024*
- MashupStack – Python *Jun 2025*
- AccelerteX – Prompt Engineering *Nov 2024*